Weather Driven Business Intelligence Platform

Technical Project Report

Document purpose	A clear and evidence based explanation of the project architecture methods results and limitations
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Weather driven decisions explained with transparent assumptions and reproducible analysis

Executive summary

This project demonstrates how weather data can be combined with business operations data to estimate and communicate weather related commercial effects. It joins two years of synthetic hourly weather and business telemetry across four United States cities, controls for recurring calendar patterns, and presents the results through an interactive Streamlit dashboard. Live weather forecasts enrich the demonstration, while uploaded merchant data can be used to estimate business specific relationships.

The principal conclusion is that the platform is a strong end to end analytical prototype. Its value lies in data alignment, transparent feature engineering, adjusted regression, scenario exploration, and accessible visualization. It should not be presented as a production forecasting service or as proof of causal business effects because the principal historical dataset is synthetic and the forward revenue projections are rule based.

Project dimension	Current implementation
Historical data	Synthetic weather and cross sector operational telemetry for 2023 and 2024
Geographic scope	New York City Chicago Seattle and Phoenix
Processed warehouse	70 045 synchronized city hour observations and 61 variables
Analytical scope	21 operational metrics across six economic sectors
Live enrichment	Open Meteo forecast data NASA POWER reference data and NASA Worldview links
Interface	Five tab Streamlit application with interactive Plotly charts
Verification	Seven automated tests passing in the reviewed environment

1 Project purpose and business question

Businesses experience weather differently. Rain can increase delivery demand while reducing restaurant footfall and bike usage. Heat can lift cold beverage sales while increasing electricity and HVAC demand. The platform turns these relationships into a portfolio view that asks three practical questions:

1. How do weather conditions relate to revenue and operational demand after recurring calendar patterns are considered
2. How might one weather event propagate across connected sectors such as mobility dining delivery and freight
3. What staffing inventory and operating actions should managers consider before forecast weather arrives

The system is designed as a demonstration of analytical workflow rather than a substitute for a company forecasting process. Its scenarios help users explore assumptions and prioritize questions for real data collection.

2 System architecture

1 Generate raw feeds	2 Align time and location	3 Engineer features	4 Estimate relationships	5 Simulate impacts	6 Present decisions

The architecture separates data creation, processing, modeling, business logic, external weather retrieval, and presentation. This modular structure makes the project easier to test and extend than a single notebook or monolithic dashboard.

Component	Responsibility	Primary output
data generator	Creates reproducible synthetic weather and operational records with seasonality and weather sensitivity	Raw CSV feeds
data pipeline	Normalizes timestamps resamples business events imputes weather gaps and engineers analytical variables	Merged Parquet warehouse
econometrics engine	Fits adjusted regression models and evaluates user defined weather scenarios	Coefficients comparisons and scenario metrics
revenue analytics	Applies business catalog assumptions and creates portfolio ripple narratives	Revenue series and operational guidance
live weather	Retrieves current and forecast conditions with resilient fallback behavior	Seven day weather package
Streamlit application	Coordinates controls charts diagnostics and explanations	Interactive dashboard

3 Data generation and integration

3.1 Source design

The historical analysis uses two intentionally mismatched synthetic feeds. Weather observations arrive hourly in UTC with sensor gaps. Business events arrive every 15 minutes in local time and contain operational measures for delivery mobility retail hospitality energy and freight. The mismatch creates a realistic engineering exercise without implying that the records came from real merchants.

Raw source	Rows	Selected fields	Timing
Weather telemetry	69 106	Temperature precipitation snow wind humidity and condition	Hourly UTC
Cross sector business stream	279 249	Orders mobility retail dining energy freight and warehouse measures	15 minute local time

3.2 Alignment and feature engineering

- Local business timestamps are assigned the appropriate IANA timezone and converted to UTC.
- Fifteen minute business records are aggregated into hourly measures using sums or means according to the metric.
- Continuous weather gaps are filled using time based interpolation; missing precipitation and snow increments are set to zero.
- The weather and business tables are joined by city and UTC hour.
- Calendar controls include hour weekday month weekend and selected holiday indicators.

- Weather features include rain categories temperature categories one and two hour precipitation lags and temperature anomalies.
- Cross sector ratios measure courier utilization delivery versus dine in activity and rideshare versus bike usage.

The resulting warehouse contains 70 045 observations with no missing fields and no duplicated city hour keys in the reviewed output. Temperature anomaly is calculated relative to the average for the same city month and hour in the available dataset.

4 Analytical method

4.1 Why raw correlation is insufficient

Weather and commerce both follow recurring cycles. A rainy Friday evening can coincide with high delivery demand even when Friday itself explains much of the increase. Comparing raw correlation with an adjusted model shows why calendar patterns should be considered before attributing an operational change to weather.

4.2 Regression specification

For each target metric the project fits an ordinary least squares model containing precipitation snow temperature anomaly wind and categorical controls for hour weekday month holiday status and city. HC1 heteroskedasticity robust standard errors are used. The implementation is best described as OLS with city and calendar fixed effects.

Interpretation. A precipitation coefficient estimates the average within model change in the target associated with one additional millimeter of precipitation while the included controls are held constant. Percentage effects are created by dividing the coefficient by the target mean.

Important limitation. These estimates are adjusted associations. The model does not by itself prove causality and the synthetic generator already embeds many of the relationships the regression later recovers.

Reported quantity	What it means	What it does not mean
Coefficient	Estimated conditional change in the target for a one unit change in a weather variable	Guaranteed causal effect
P value	Compatibility of the estimate with a null hypothesis under the model assumptions	Probability that the effect is true
R squared	Share of in sample variation explained by the fitted model	Out of sample prediction accuracy
Scenario delta	Difference between two model predictions under specified inputs	Observed future revenue

5 Business sectors and scenario logic

Sector	Illustrative metrics	Expected weather pathway
Delivery and logistics	Orders couriers surge delivery time cancellations	Rain can increase demand while reducing available courier supply
Urban mobility	Bike trips transit rideshare congestion	Travelers substitute away from exposed transport modes

Sector	Illustrative metrics	Expected weather pathway
Retail and commerce	Footfall rainwear winter apparel ecommerce	Weather changes store visits product mix and online demand
Hospitality	Dine in guests patio use hot food cold beverages	Conditions alter venue choice and product preference
Energy	Grid load and HVAC demand	Temperature extremes increase heating or cooling requirements
Freight	Road delay and warehouse dispatch	Poor conditions can reduce transport reliability and throughput

The dashboard converts selected weather conditions into sector predictions and business specific revenue illustrations. Domino chains describe plausible propagation paths such as rain reducing bike use increasing rideshare demand worsening congestion and extending delivery times. These chains are useful scenario narratives but are not estimated structural causal networks.

6 Dashboard experience

Dashboard view	Purpose
Weekly Operational Impact	Compares modeled actual revenue with a clear weather baseline for the selected portfolio
Historical Trends and Seasonality	Shows longer term revenue patterns and compares raw correlations with adjusted model estimates
Business Interactions and Domino Chains	Explains cross sector substitutions thresholds and downstream operational effects
Live Scenario Simulator and Guidance	Uses current or user selected weather conditions and displays a seven day weather adjusted projection
Data Wrangling Diagnostics	Exposes alignment logic data volumes engineered variables and synchronized records

Users may choose preset businesses create a custom category or upload CSV or Excel revenue history. Custom business parameters are inferred from keywords or supplied directly. Uploaded merchant revenue is aligned with the project weather history to estimate business specific sensitivities when enough overlapping dates exist.

7 Live data and forecasting boundaries

- Open Meteo supplies the current conditions and seven day meteorological forecast used by the application.
- NASA POWER supplies a recent historical reference record rather than a same moment confirmation of current conditions.
- NASA Worldview is presented as an external imagery browser link and is not analyzed automatically by the model.
- Square and Stripe connector classes can call external APIs when credentials are supplied, but the current application initializes them without credentials and therefore uses calibrated sandbox sequences.
- Seven day revenue projections apply configured elasticities and weekday weights to weather forecasts. They are deterministic scenario projections rather than trained machine learning or generative AI forecasts.

8 Verification and current quality

The project includes unit and integration checks for data generation pipeline alignment scenario output connector schemas CSV column detection and merchant elasticity estimation. All seven tests passed during this review. The processed warehouse was also checked for missing values and duplicate city hour keys.

Verified item	Result
Automated tests	7 passed
Processed observations	70 045
Processed variables	61
Missing values	0 in the processed warehouse
Duplicate city hour keys	0
Historical period	1 January 2023 through 31 December 2024 subject to city timezone boundaries

9 Limitations and recommended next steps

Priority	Current limitation	Recommended improvement
High	Historical relationships are generated rather than observed	Validate with permissioned merchant and operational data from multiple businesses
High	Adjusted regressions are described too strongly as causal	Use careful association language and develop a credible identification strategy before causal claims
High	Default Square and Stripe records are simulated	Add explicit credential configuration status labels pagination and visible connector errors
High	Merchant model fit is constrained to a preset range	Report the actual R squared and add standard errors confidence intervals and holdout validation
Medium	Seven day revenue projections are deterministic	Add backtesting uncertainty intervals and benchmark models
Medium	Weather sources represent different observation times	Display source timestamps prominently and avoid treating delayed NASA data as a current consensus check
Medium	Dashboard supports four selectable cities	Add geocoding and configurable city metadata if broader coverage is required

10 Conclusion

The Weather Driven Business Intelligence Platform is a coherent demonstration of multi source data engineering econometric adjustment scenario analysis and executive visualization. It succeeds as a portfolio project because the complete analytical path is visible from raw data through operational recommendations. Its credibility is strongest when the synthetic nature of the history and the rule based nature of the projections are stated directly.

The most valuable next step is not additional visual complexity. It is validation with real revenue and operational records followed by honest out of sample testing. That would convert the current demonstration into a defensible business forecasting product.

Overall assessment	Finding
Portfolio value	Strong demonstration of a complete data and analytics workflow
Technical maturity	Well structured prototype with tested core components
Evidence maturity	Requires validation on observed merchant and operational data
Recommended positioning	Decision support demonstration rather than production causal forecasting

References and project files

- Project README and portfolio case study
- src data_generator.py data_pipeline.py econometrics.py revenue_analytics.py live_weather.py and revenue_connectors.py
- tests test_pipeline.py and test_connectors.py
- Open Meteo Weather Forecast API documentation
- NASA POWER Hourly API documentation
- NASA Worldview Earth observation browser